mircean2 at CheckThat! 2026: Reasoning-Path Verification and Reranking for Numerical Claim Fact-Checking*

Multilingual Verification Using Large Language Models

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Abstract

This paper presents the mircean2 team's approach to the CheckThat! 2026 shared task on fact-checking numerical claims. We address the multilingual Track 2 challenge, verifying numerical claims in Arabic, English, and Spanish. Our methodology leverages fine-tuned Large Language Models enhanced with dynamic question generation for evidence-based verification. For Arabic, we employ a Qwen2.5-1.5B-Instruct model with Camel-based preprocessing and sigmoid-weighted verification pipelines, achieving a macro F1 score of 0.8602. For English and Spanish, we utilize Llama 3.2 1B models adapted for numerical claim verification. Our results demonstrate strong performance across all three languages, with particular success in verifying true claims and handling conflicting evidence.

Keywords

fact-checking, numerical claims, multilingual NLP, large language models, question generation, CheckThat! 2026

1. Introduction

Fact-checking has emerged as a critical challenge in combating misinformation. While most research focuses on binary veracity classification, the verification of numerical claims presents distinct challenges, requiring systems to understand not just semantic content but also quantitative accuracy. The CheckThat! 2026 shared task extends this challenge to the multilingual setting, requiring systems to verify numerical claims across Arabic, English, and Spanish—languages with diverse linguistic properties and minimal shared resources.

This paper describes our participation in Track 2 of CheckThat! 2026, where we developed a multilingual fact-checking system specifically tailored for numerical claims. Our approach leverages state-of-the-art Large Language Models (LLMs) with language-specific adaptations and dynamic evidence generation, achieving competitive results across all three target languages.

2. Task Description

The CheckThat! 2026 Track 2 task requires systems to classify numerical claims into one of three categories: *True*, *False*, or *Conflicting*. The key challenge lies in handling claims where:

- Numerical values must be verified against retrieved evidence
- Claims span multiple languages with distinct linguistic and cultural contexts
- Evidence may partially support or contradict claims (conflicting category)
- Arabic claims require specialized preprocessing to handle diacritics and morphological complexity

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3. Approach

3.1. System Architecture

Our system follows a three-stage pipeline:

1. **Evidence Generation**: Dynamic question generation from claims to retrieve supporting evidence
2. **Encoding**: Joint encoding of claims, questions, evidence, and verdicts
3. **Classification**: Fine-tuned neural classifiers for three-way label prediction on reasoning traces

3.2. Language-Specific Models

3.2.1. Arabic

For Arabic numerical claims, we employ a Qwen2.5-1.5B-Instruct model with LoRA fine-tuning. The pipeline incorporates:

- Camel-based Arabic preprocessing to normalize text and handle morphological variations
- Sigmoid-weighted scoring to balance evidence strength across multiple sources
- Context-aware question generation specifically targeting numerical properties

3.2.2. English and Spanish

For English and Spanish, we utilize Llama 3.2 1B models adapted through LoRA-based parameter-efficient fine-tuning. This choice prioritizes efficiency while maintaining competitive performance. Both languages share a similar pipeline with:

- Standard text preprocessing and tokenization
- Fine-tuning on task-specific claim-evidence pairs

3.3. Training Details

All models were fine-tuned on Google Colaboratory A100 GPUs with a total computational budget of 40 EUR. We employed:

- LoRA rank: 16
- Learning rate: 2e-4
- Batch size: 32 per language
- Maximum input length: 512 tokens

4. Results

4.1. Evaluation Metrics

We report results using the official task metrics:

- **Average Score**: Weighted average across all claims
- **Macro-averaged F1**: Unweighted mean F1 across all three classes
- **Recall@5**: Recall within top-5 predictions
- **Class-specific F1**: Per-class performance metrics

Language	Avg Score	Macro F1	Recall@5	True F1	Conflict F1	False F1
Arabic	0.5992	0.8602	0.3382	0.8345	0.0	0.886
Spanish	0.4778	0.6607	0.2949	0.5742	0.5017	0.9061
English	0.4180	0.6010	0.2350	0.6998	0.2265	0.8767

Table 1
Performance across languages. Arabic achieves the highest macro-averaged F1 score (0.8602), while Spanish and English show competitive results in false claim detection (F1 > 0.90).

5. Analysis

5.1. Strengths

The results demonstrate several key findings:

- **Arabic Performance**: The Qwen2.5-1.5B model combined with Camel preprocessing yields the strongest overall performance (macro F1 = 0.8602). This suggests that morphologically-aware preprocessing and larger model capacity are beneficial for Arabic fact-checking, also that Qwen family models are specialized in Arabic
- **False Claim Detection**: Across all languages, the system achieves high F1 scores for false claims (0.886, 0.9061, 0.8767 for Arabic, Spanish, English respectively). This indicates the models are effective at identifying explicit contradictions with evidence.

5.2. Limitations

Several factors limit our current results:

- **Conflicting Class Imbalance**: The zero F1 score for Arabic conflicting claims suggests class imbalance or insufficient training examples for this category.
- **Recall Challenges**: Recall@5 scores below 0.35 across all languages indicate that the models struggle with evidence-based ranking, possibly due to limited training data or suboptimal question generation strategies. The highest scoring model didn't use question generation.
- **Model Capacity Trade-offs**: Using Llama 3.2 1B for English and Spanish, while computationally efficient, appears to underperform compared to the larger Arabic model, suggesting that model scale impacts multilingual numerical reasoning.

6. Perspectives for Future Work

Future improvements to our system could address several dimensions:

1. **Enhanced Question Generation**: Implementing multi-hop reasoning in question generation could improve evidence retrieval by decomposing complex numerical relationships.
2. **Balanced Training Data**: Collecting or augmenting data for underrepresented classes (particularly "Conflicting") would help address the observed class imbalance.
3. **Cross-lingual Transfer**: Investigating zero-shot and few-shot transfer from high-resource to low-resource languages could improve English and Spanish performance.
4. **Ensemble Methods**: Combining predictions from multiple models or approaches could boost robustness and reduce variance in performance.
5. **Numerical Reasoning Modules**: Integrating specialized modules for numerical comparison and unit conversion could enhance accuracy on technical claims.
6. **Adversarial Evaluation**: Testing on adversarially perturbed claims and evidence would reveal model robustness and guide further improvements.

7. Conclusion

Our multilingual fact-checking system demonstrates competitive performance on the CheckThat! 2026 Track 2 task across Arabic, English, and Spanish. The approach leverages language-specific model choices and preprocessing strategies, achieving a macro-averaged F1 of 0.8602 for Arabic, 0.6607 for Spanish, and 0.6010 for English. While our results highlight strengths in false claim detection, areas for improvement remain in handling conflicting evidence and optimizing evidence-based ranking. Future work will focus on addressing class imbalance, improving question generation, and exploring cross-lingual transfer learning.

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Declaration on Generative AI

The authors have not employed any Generative AI tools in the research or writing of this paper. All experiments were designed and implemented by the authors.

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